Reg. No. | | | | | | | | | | |

[18DSOE04]

SEVENTH-SEMESTER B.Tech EXAMINATION, DECEMBER-2021

BIG DATA ANALYTICS

Time: 3 Hours **Maximum Marks: 70**

Instructions:
i. Answer **FIVE** full Questions. Question number 1 and 2 are compulsory. Each question carries 14 marks.
ii. Missing data may be suitably assumed.

1.	**a.**	Define big data. Describe the features of big data that separates it from the traditional datasets.	**07**
	b.	Illustrate read and write operation in HDFS with a suitable diagram.	**07**
2.	**a.**	With a code snippet of word count from a document dataset, explain the use of mapper and reducer in a MapReduce program in obtaining efficiency than the traditional programs.	**07**
	b.	Draw YARN architecture and explain the difference between Hadoop 1.x and 2.x Architecture.	**07**
3.	**a.**	Explain atomic and complex data types of Apache Pig with an example of each.	**06**
	b.	Describe the data processing of Apache pig with a simple example while using the following operations: Load data, Filter data, Store the results data, Display the result, Check the filters.	**08**

OR

4.	**a.**	With code explain 3 bag and tuple functions and 5 relational operators of Pig.	**08**
	b.	Write a Pig script which load a HDFS text file content into pig while separating the file using delimiter "," and using a schema. Explain the script as well.	**06**
5.	**a.**	Discuss the key features of Hive. Compare Hive with respect to Pig.	**4+4**
	b.	Explain Hive architecture in detail with a diagram.	**07**

OR

6.	**a.**	Write down Hive commands for achieving the following task:	**09**

i. Create an internal table named "employee" with 4 columns with comments for each columns and a table property.
ii. Load any file content to the table "employee".
iii. Change the table name to "employee_data".
iv. Add a new column to the table.
v. Rename an existing table.
vi. Drop an existing column.

	b.	Write down a short note on static and dynamic partitioning of Hive tables with an example to each.	05
7.	a.	Explain the usage of Apache Sqoop. Explain the Sqoop working with a neat diagram.	06
	b.	Write a detailed note on working of Sqoop import and role of where clause in this aspect with an example.	08

OR

8.	a.	What is Apache flume? Describe the features and components of the Apache flume in detail.	08
	b.	Illustrate the working of the Apache flume in detail with an example.	06